



Final Report

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CMPSC 185 W26 - TEAM02



Project Name & Value Proposition

The project, **Frame**, is designed around the idea that *memories shouldn't be work*. Frame focuses on simplifying how people collect and share photos after shared experiences such as trips, events, or gatherings. Today, photos are captured easily but become difficult to organize and distribute afterward, especially across different device ecosystems. Frame's value proposition is **"Sharing memories made simple."** By automatically collecting, curating, and synchronizing photos into shared albums across groups, Frame transforms scattered images into organized memories that can be easily accessed and enjoyed by everyone involved.

Team Members & Roles

Our team consisted of three members who contributed to different aspects of the design process.

Gavin Cao contributed primarily to early ideation and concept development, helping explore how people currently share photos and how that experience could be improved through design. Rachit Gupta focused on implementation and technical considerations, helping translate conceptual designs into medium- and high-fidelity interface prototypes. Zeel Patel contributed heavily to design synthesis, task flow development, and user research documentation while also supporting the development of visual prototypes and usability evaluations.

Each member participated in interviews, brainstorming, and testing sessions throughout the quarter, ensuring that design decisions were collaborative and grounded in user insights.



Problem / Solution Overview

Problem.

Photo sharing often breaks down after the moment when the photos are taken. Instead of being able to simply enjoy their memories, users are forced into manual tasks such as sorting photos, switching between apps, and repeatedly following up with friends to collect missing images. This process becomes even more complicated when groups include both iOS and Android users. In many cases, one person becomes the “default sharer,” acting as a bridge between platforms by gathering and redistributing photos to everyone else. Over time, this role demands significant time and effort, turning what should be a joyful experience into a logistical responsibility. Users also report losing confidence once photos are shared, as they become uncertain about who has access to their images, how long they will remain available, and whether their boundaries around privacy are respected.

Solution.

Frame addresses these challenges through a cross-platform photo sharing system designed to make memory sharing effortless after events. The system automatically detects and clusters photos taken during the same event and organizes them into unified shared albums. Using intelligent event clustering, privacy-focused facial detection, and cross-platform synchronization, Frame collects relevant photos from participants and prepares them for intentional sharing. Users can then review and distribute albums to the appropriate audience, ensuring that photos reach the right people while maintaining control over visibility. By removing the need for manual sorting and eliminating platform barriers, Frame allows groups to share memories seamlessly across iOS and Android without relying on a single individual to manage the entire process.



Needfinding Interviews

Interview Process

To better understand how people currently capture, organize, and share photos after events, we conducted interviews with six participants representing different user types and device ecosystems. These participants included individuals who frequently organize and distribute photos, users who rarely share photos, and people who operate across different platforms such as iOS and Android. We asked open-ended questions about the last time they shared photos from a group event, how they currently distribute images, and what frustrations arise during that process. Through these conversations we began to understand that the act of taking photos is effortless, but the process of organizing and sharing them afterward is often tedious and fragmented.

Interview - Purple

We met “**Purple**,” a Gen Z user who regularly switches between both iOS and Android devices. Purple frequently attends social events with friends where many people take photos throughout the night. However, after events end, Purple often struggles to gather those photos into one place.

We were surprised to learn that Purple often receives photos through multiple channels such as messaging apps, social media, and cloud links, which makes it difficult to locate images later. Instead of having one organized album, the photos from a single event are often scattered across several platforms.

This interview revealed how fragmented the photo sharing experience can become when users rely on multiple devices and services. Purple’s experience helped us recognize that cross-platform compatibility plays a major role in how memories are shared and preserved.

Interview - Red

We met “**Red**,” a millennial Android user who regularly captures photos during trips and social gatherings. Red enjoys taking photos but rarely feels motivated to organize them afterward.



During our conversation, Red described having thousands of photos stored on their device, many of which they rarely revisit. Photos from group events often remain mixed with unrelated images in the gallery, making it difficult to find meaningful memories later.

Red's experience highlighted the challenge of long-term photo organization. Even when users capture valuable memories, those moments often become buried within large personal photo collections.

Interview - Green

We interviewed "**Green**," a Gen Z user who primarily uses iOS devices and rarely shares photos after events. Green enjoys capturing photos but prefers keeping them private rather than distributing them widely.

During the interview, Green explained that they feel comfortable with their current setup and have little motivation to actively organize or share photos. They rely on iCloud automation and iMessage for any sharing that does occur, and do not engage much with shared albums. When asked about sharing with others, Green expressed a general lack of interest rather than active reluctance, noting that the effort of organizing photos for others felt unnecessary.

Green's perspective helped us understand that not all users are motivated by the same sharing goals. While many users want to share memories with friends, some are content with passive, low-effort workflows and would only engage with sharing tools if the process required virtually no additional work.

Interview - Pink

One of the most insightful interviews came from "**Pink**," a college junior studying pre-med biology who frequently becomes responsible for sharing photos among friends and family. Pink acts as a manual link between incompatible ecosystems, juggling AirDrop for her iOS college friends and WhatsApp or Google Photos links for her Android-using family overseas.

We were surprised to hear how much effort Pink invests in maintaining photo quality and accessibility. Pink often sends photos through several different apps in order to accommodate both iOS and Android users while avoiding compression from messaging platforms.



Pink described this process as exhausting but necessary, because they want to ensure that everyone receives the full-quality images. This interview revealed the concept of the “**default sharer**,” a person who becomes responsible for managing photos across multiple platforms.

Interview - Blue

We interviewed “**Blue**,” a 21-year-old college senior studying communications who regularly attends social gatherings where many people take photos. Blue takes photos regularly but shares them selectively.

Blue explained that they often hesitate to share photos because they are unsure who will ultimately have access to them or how those images might be redistributed. Blue actively curates which photos to keep and which to share, viewing curation as a way to protect friends' privacy. Blue equates sharing with losing control, which leads them to keep their most meaningful memories offline.

This interview highlighted how privacy concerns can significantly limit sharing behavior. Blue's experience showed us that some users need reassurance that their photos will remain within intended boundaries before they feel comfortable sharing at all.

Interview - Yellow

We met “**Yellow**,” a 26-year-old marketing professional who does photography on the side. Yellow's friends and family treat him as “the photographer” of the group, and he frequently takes photos but prefers sharing only a carefully selected subset with others.

Yellow explained that sorting through hundreds of photos can feel overwhelming, and as a result they often delay sharing photos altogether. Messaging apps also make it difficult to distribute organized albums rather than individual images.

Yellow's experience emphasized the importance of curation and meaningful sharing. Many users want tools that help them highlight the best moments from an event without spending excessive time organizing photos.



POVs & Experience Prototypes

After our needfinding process, we synthesized our key findings into **point-of-view statements (POVs)** to better understand the needs of our users. These POVs helped us identify the motivations, frustrations, and opportunities present within the current photo-sharing ecosystem. The interviewees we wrote POVs for were Purple, Red, Green, Pink, Blue, and Yellow.

Ms. Purple

We met ms. purple, a student who often becomes responsible for taking and sharing photos after group trips and event.

We were surprised to notice that even though sharing photos across multiple apps is exhausting and confusing, she still takes on the responsibility of making sure everyone gets the memories.

We wonder if this means she values preserving shared experiences more than her own convenience.

It would be game changing to support people in sharing group memories without constant manual effort or mental tracking.

HMWs from Purple: How might we...

- allow photos to organize themselves by event, people, or moment?
- centralize group memories while still respecting people's preferred platforms?
- **make sharing photos feel like a background action?**
- help people feel confident that everyone has received the photos?
- enable photos to be shared across platforms automatically?

Mr. Blue

We met mr. blue, a 21 year old who is hesitant to share photos widely because they are concerned about privacy and long-term access.

We were surprised to notice that their discomfort comes from not knowing who will see the photos later or how they might be reused, not from the platform/service.



We wonder if this means they want to participate in shared memories while still maintaining control over visibility and access.

It would be game changing to help people share photos in ways that respect personal boundaries and control.

HMWs from Blue: How might we...

- How might we help people stay aware of who their photos are shared with over time?
- **let people share meaningful photos without giving up control over who sees them later?**
- reduce the anxiety of not knowing where shared photos will end up?
- make photo sharing feel reversible?
- make it clear who can see a photo before it's shared?

Mr. Yellow

We met mr. yellow, a professional photographer who regularly manages and shares large sets of photos with different people.

We were surprised to notice that even with expertise, sorting and delivering relevant photos still requires a lot of manual work.

We wonder if this means they need better support in surfacing the right photos for different audiences.

It would be game changing to make it easier to organize and present meaningful photos without repetitive manual curation.

HMWs from Yellow: How might we...

- prevent photographers from choosing between speed and quality?
- make it easier to organize photos by people rather than dates or folders?
- help photographers ensure clients see what matters most to them?
- reduce the repetitive manual work?
- **help photographers find the right photos for people without re- sorting everything each time?**



Top 3 Solutions & Experience Prototypes

After brainstorming solutions inspired by our HMW questions, we selected three ideas to explore further through **experience prototypes**. Each prototype tested a different assumption about how automation, awareness, and organization could improve the photo sharing experience.

1. Background Photo Sharing

Inspired by Purple - HMW: “How might we make sharing photos feel like a background action?”

The idea behind this solution is that the app automatically curates and shares photos with the right people in the background using contextual signals such as event timing, location, and people present in the images.

Experience Prototype 1

The core assumption tested was that people would feel less anxious and more trusting about sharing photos if they could simply drop all their photos into the system and allow it to automatically organize and distribute them.

For the setup, participants were given a stack of printed photos representing images taken across multiple events and involving different people. They were also shown labeled audience cards such as **Friends, Family, Event Group, and Individual Person**.

Participants handed the stack of photos to the “app” (represented by a team member). This simulated the idea that the app syncs with the user’s photo library. Without further participant input, the facilitator automatically grouped photos into event-based piles and assigned audiences based on content. For example, group photos were shared with friends or family groups, while photos containing a specific person were assigned to that person’s library.

The grouped piles were then placed in front of the participant as visible “albums,” representing how the app would automatically organize memories.

Participants were then asked to reflect on whether the automatic grouping matched their expectations, whether they trusted the system to send photos to the correct audience, and whether this workflow reduced effort compared to manual sharing.



What worked was that participants found the “**drop-and-done**” interaction intuitive, and many appreciated that automatic grouping reduced decision fatigue. However, some participants expressed a desire for a quick confirmation of who received which photos.

The key implication

was that **one-time automated sharing shows strong promise**, but the system should provide **light transparency without introducing additional work**.

2. Visible Audience Sharing

Inspired by Blue - HMW: “How might we help people stay aware of who their photos are shared with over time?”

This solution focused on making the intended audience visible throughout the sharing process so that users maintain confidence about where their photos are going.

Experience Prototype 2

The core assumption tested was that people would feel more confident sharing photos if the intended audience remained visible during the sharing process.

Participants were given a stack of printed photos from a shared event along with labeled audience groups such as **Friends, Family, Event Group, and Private**.

For each photo, participants handed the image to the “app.” The facilitator then presented a suggested audience based on the content of the photo. The suggested audience label remained visibly attached to the photo while the participant decided whether to confirm or change the recipient group.

Participants were then asked to reflect on whether seeing the audience upfront influenced their decisions and whether the visibility reduced anxiety about accidentally sharing photos with the wrong people.

What worked was that visible audiences made sharing feel **more intentional**, and many participants reported reduced anxiety around “who will see this.” AI-generated audience suggestions also reduced mental effort.

However, some participants felt that repeated confirmation steps created too much friction when sharing many photos.



The key implication

was that **awareness builds confidence**, but repeated confirmations reduce momentum. Future versions of the design should surface audience information once rather than repeatedly asking for confirmation.

3. Professional Photo Organization Mode

Inspired by Yellow - HMW: “How might we help photographers find the right photos for people without re-sorting everything each time?”

This solution explored a professional-style workflow where photos are automatically organized into event-based projects that photographers can quickly review and share.

Experience Prototype 3

The core assumption tested was that automatically grouping photos into date-based event projects would reduce the need for photographers to repeatedly reorganize large collections.

Participants were given a messy stack of printed photos representing images from one event. They were also provided with a sticky note representing the event name and time, along with a table that simulated a workspace.

Participants dropped all photos onto the table at once, representing uploading them to the system. The facilitator then immediately grouped the photos into a single pile under the event label, simulating automatic event clustering.

Participants were then asked to reflect on whether skipping manual sorting felt relieving or unsettling, whether they trusted the automated grouping, and whether the workflow felt faster without sacrificing control.

What worked was that **date-based grouping removed the need to rethink structure**, and participants appreciated the feeling that the event album was immediately “project ready.”

However, some users expressed concern that fully automated grouping removed too much control.

The key implication

was that while users appreciate removing repetitive sorting work, they still want a **lightweight review moment before sharing** to confirm the system’s decisions.



Design Evolution

Final Solution

Description

Frame's final solution is a cross-platform photo-sharing experience designed to keep memories effortless after the moment ends. It removes the burden of manual sorting, app-switching, and follow-ups by automatically creating shared albums and making audience visibility intentional and clear.

Instead of turning one person into the "default sharer," Frame helps groups share photos seamlessly across iOS and Android without losing trust or time. The system combines intelligent event clustering, privacy-focused facial detection, and cross-platform synchronization to collect relevant photos from participants and prepare them for intentional sharing.

Rationale for the Selected Solution

Our final solution was grounded in evidence gathered across all three experience prototypes. From Background Photo Sharing, we learned that one-time automated sharing shows strong promise, but the system should provide light transparency without introducing additional work. From Visible Audience Sharing, we found that awareness builds confidence, but repeated confirmations reduce momentum, meaning the design should surface audience information once rather than repeatedly asking for confirmation. From Professional Photo Organization, we learned that while users appreciate removing repetitive sorting work, they still want a lightweight review moment before sharing to confirm the system's decisions. These findings led us to a design that combines automation with intentional review: Frame automatically clusters and organizes photos into albums, but always gives users a lightweight moment to accept, decline, or edit before anything is shared.



Tasks

Simple Task

The user receives a shared album and accepts or declines it. This task represents the most basic interaction a user would have with Frame. The user receives a notification that an album has been shared to them, views who sent it along with the most relevant photos and other photos of the album, and then chooses to accept or decline. This task is important to our user base because it addresses the need discovered through our interview with Blue, where users wanted to feel confident about who has access to their photos and maintain control over what enters their personal library.



Image 1: Simple task flow



Moderate Task

Moderate Task: User discovers automatically curated albums by Frame. In this task, the user receives a notification once an album is automatically created and has options to choose from for the album. The user can review and edit the album by adding or removing photos, changing the timeline, and information like the name of the album, and can add more people from the suggested contacts or by searching for them by name. In this instance, the user edits the auto-curated album. This task is important because it directly responds to the needs of users like Purple and Yellow, who expressed frustration with the manual effort required to organize and share photos after events. Automatic album curation reduces that burden significantly.



Image 2: Moderate task flow



Complex Task

User imports all or selected photos from their gallery into the app. This task represents the initial setup process where the user decides how much access to grant Frame. The user navigates to the settings pop-up page, chooses to import all photos or selected photos, selects the photos they want the app to have access to, and then sees the selected photos on the dashboard. The user can also change permissions later if they initially did not allow access. This task is important because it addresses the privacy concerns raised by Green, who hesitated to share photos due to uncertainty about access and control, as well as the onboarding friction that was a major concern in our heuristic evaluation.



Image 3: Complex task flow



Design Evolution Visualization and Rationale

Description

Our design evolved through four major iterations: concept sketches, paper prototypes, medium-fidelity wireframes, and the final high-fidelity prototype.

Concept Sketches → Paper Prototypes. Our earliest concept sketches explored a range of platforms, including a mobile phone app, a desktop collaborative album editor, and AR glasses for viewing location-based memories.

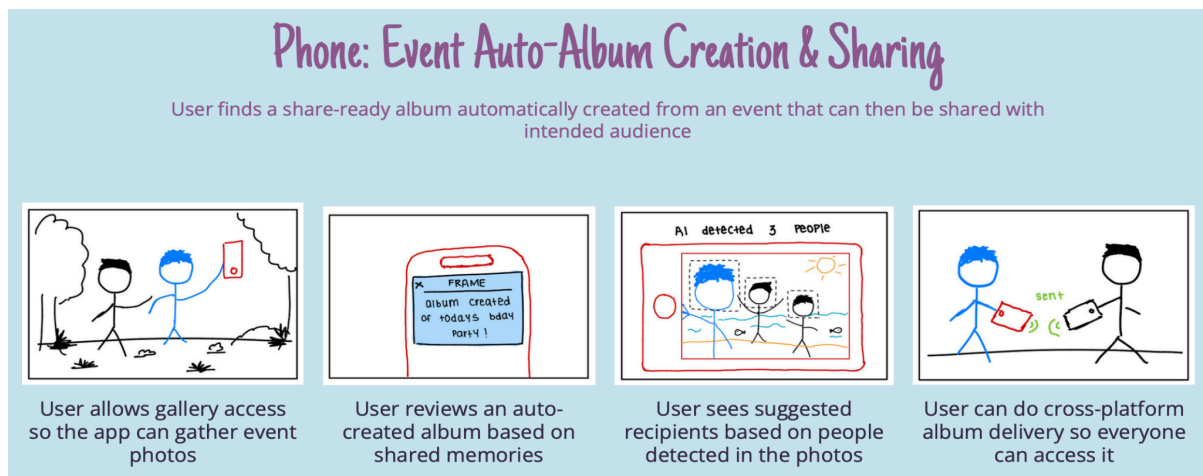


Image 4: Concept sketch for what Frame would look like on a mobile phone

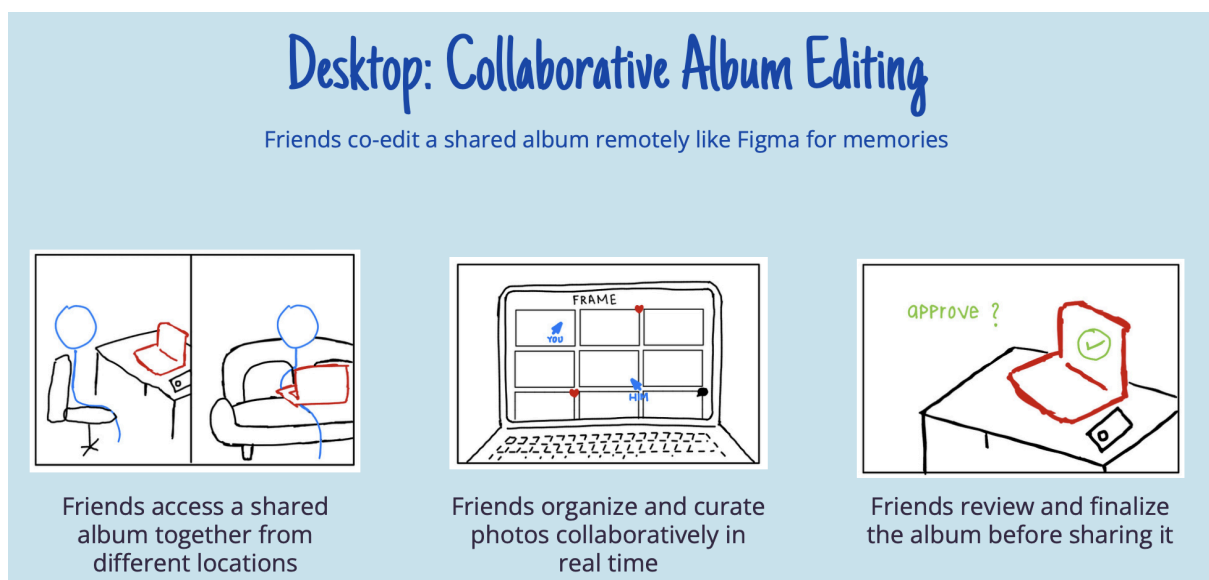


Image 5: Concept sketch for what Frame would look like on a desktop

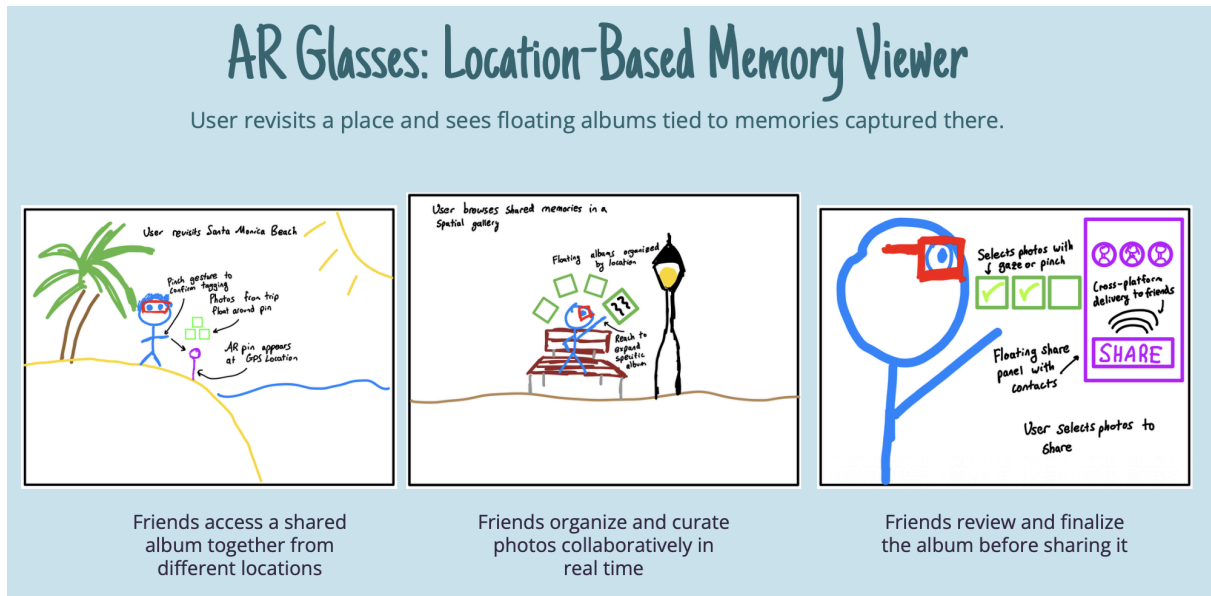


Image 6: Concept sketch for what Frame would look like through AR glasses

We selected the mobile phone concept as the most realistic and impactful platform for our users and proceeded to construct a paper prototype. Prototype interfaces were first sketched collaboratively on iPads, then transferred to paper and printed/cut out to create consistent low-fidelity screens.

Key flows such as gallery import, auto-album creation, shared album notifications, and album editing were constructed, along with supporting modals for permission prompts, sync screens, participant editing, timeline adjustments, and keyboard inputs.

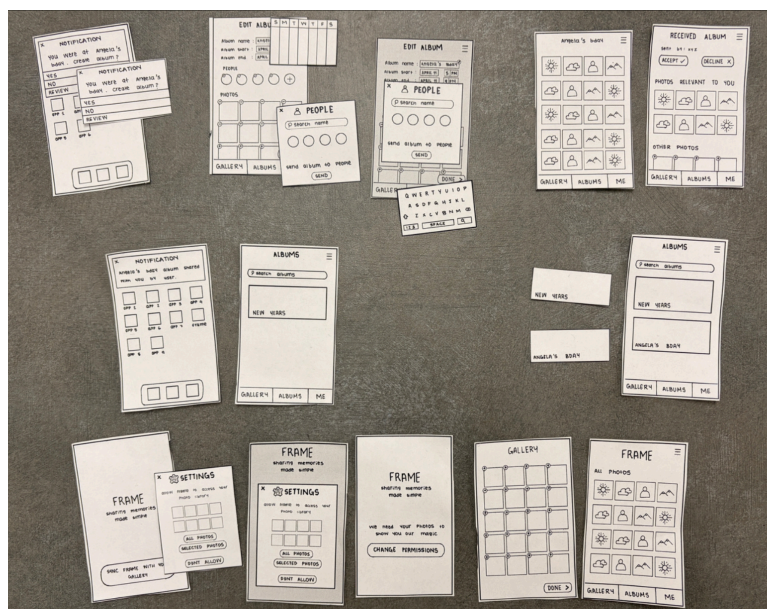


Image 7: Cutouts



During testing, one team member acted as the "computer," manually swapping screens and placing interface components in response to user actions.

Paper Prototypes → Medium-Fidelity Wireframes

We tested the paper prototype with three participants using a "Think Aloud" protocol in quiet settings. We measured usability through misclicks, task completion time, and mental effort. Results showed that the notification flow worked extremely well. All users instantly understood the shared album notification, and the Accept/Decline simple task had a 100% success rate with zero confusion. Users also showed a high value perception of auto-creation. One participant noted how much time it saved compared to manually picking photos. Users also liked seeing faces (avatars) on the album cover before opening it. Based on these results and the interaction confusion found during testing, we moved to medium-fidelity wireframes in Figma. Key changes included replacing the ambiguous timeline text boxes with standard, native iOS-style scrolling date/time pickers to fix interaction confusion found in the moderate task. Our two usability goals for this stage were learnability (measured by misclick rate) and efficiency (measured by time on task).

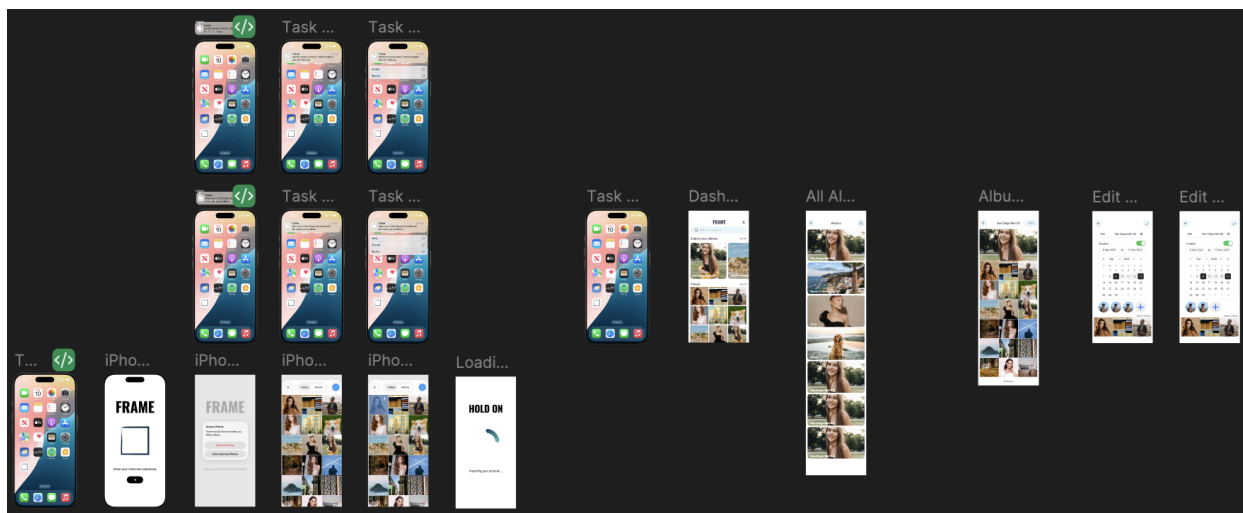


Image 8: Frame Medium-fidelity prototype

A link to our working medium-fidelity prototype can be found [here](#)



Medium-Fidelity → High-Fidelity Prototype: Heuristic Evaluation

Our medium-fidelity prototype was evaluated by external evaluators using Nielsen's 10 usability heuristics.

The evaluation identified a total of 75 violations across all heuristics.

The biggest pain points were

H1: Visibility of system status (13 violations),

H3: User control and freedom (10 violations),

H4: Consistency and standards (9 violations),

H5: Error prevention (9 violations), and

H6: Recognition rather than recall (9 violations).

By severity, there were 16 severity 1 violations, 30 severity 2, 22 severity 3, and 7 severity 4.

The four most common categories of requested changes were:

1. Add Recovery and Confirmation for Destructive Actions: Evaluators flagged that discarding albums or declining invitations had no confirmation dialogs, and there was no undo for accidental taps. We addressed this by implementing a 30-day "Recently Discarded" bin mirroring iOS Photos so users can recover accidentally dismissed or declined albums. We also added a discard changes dialog when users try to go back after making edits.

2. Improve System Feedback for Long or Background Processes: Evaluators noted that when Frame auto-creates an album in the background, the only feedback was a push notification with no in-app status indicator. We addressed this by adding a Notification indicator on the Dashboard Page and a New Album instance on the Albums Page with quick buttons to accept, decline, or click the card to review.

3. Strengthen Interaction Affordances and Recognizability: Evaluators found that avatar bubbles in the People section had no names displayed and that notification action labels were inconsistent between tasks. We created a brand new Edit Album Page with better people management, including visible name labels, and standardized notification actions across tasks to Accept, Decline, and Edit.



4. Clarify Permission and Privacy Decisions: Evaluators noted there was no onboarding tutorial or explanation of what Frame would do with photos during the initial setup. We added an extra screen with better wording and information on how Frame will use and access the photos that the app has been granted permissions for. For the "Allow All Photos" scope warning concern, this behavior is a native iOS feature rather than functionality implemented by our app, so we included a Change Permissions page that allows users to update or modify their permission settings at a later time.

For the severity 4 violations specifically, the full-screen "HOLD ON" loading screen with no progress bar was fixed by removing the loading screen entirely after researching the native iOS permissions flow. The irreversible album discard with no confirmation or undo was fixed by implementing the 30-day "Recently Discarded" bin. The "Allow All Photos" option, having no scope warning, was rejected because this is native iOS behavior, though we included the Change Permissions page as a mitigation. The issue of only certain blue-highlighted elements being clickable across the prototype was rejected because this is Figma prototyping functionality, not something implemented by our app.

For the severity 3 violations, the lack of in-app status for background album creation was fixed by adding Dashboard notifications and Albums Page new album cards. The "SYNC FRAME WITH YOUR GALLERY" using technical jargon was rejected because the medium-fidelity prototype had already replaced this with a native OS-style permissions pop-up. The missing cancel button on the import screen was fixed by removing the loading screen and including the Change Permissions page. Inconsistent notification interaction models between tasks were fixed by standardizing action labels to Accept, Decline, and Edit. Unlabeled avatar bubbles in the Edit Album People section were fixed by creating a new Edit Album Page with better people management. The missing onboarding tutorial was fixed by adding an extra screen with better wording and information. The missing Cancel/Discard Changes option on the Edit Album screen was fixed by adding a prompt dialog when users try to go back after making changes. The small pencil icon for editing the People section was rejected because this referred to the older medium-fidelity prototype. The lack of efficient photo selection shortcuts in the Selected Photos path was rejected because this interaction is controlled by the native iOS photo permission interface. The missing error message when a user denies photo access was rejected because denying access is a valid user choice and can always be changed in the Change Permissions page. The implicit KEEP behavior when selecting REVIEW was fixed by landing users on a new review page with options to Accept, Decline, and Edit.



Final Prototype

Hard-Coded Data Used

Our final high-fidelity prototype in Figma used hard-coded sample data to simulate realistic interactions. This included placeholder album names such as "Angela's Bday" and "New Year's," sample profile photos for contact avatars, placeholder photo thumbnails in album grids and the dashboard gallery, and static notification content showing album sharing prompts and event detection messages. All task flows were built using Figma's interactive prototyping features with linked screens rather than dynamic data.

AI Tools Used

We used AI tools at several points during the project with clear justification for each use case. AI was not used to generate any design decisions, user research findings, or prototype layouts. All design work was completed manually in Figma by the team.

Reflection & Next Steps

Main Learnings

This quarter taught us how much the design thinking process relies on iteration and user feedback rather than assumptions. One of our biggest takeaways was that what feels intuitive to the designer is not always intuitive to the user. The heuristic evaluation was especially valuable in revealing issues we had overlooked, such as the lack of recovery options for destructive actions and the inconsistency of notification interaction patterns across tasks. We also learned the importance of grounding design decisions in evidence from real user interactions.

Another key learning was about scope. Our early explorations included AR glasses and desktop collaborative editing concepts, but testing quickly showed that the mobile app was the most realistic and impactful platform for our users. Learning to narrow our scope based on evidence rather than ambition was an important part of this project.



Future Additions

If we had more time, there are several features we would explore adding to Frame. First, we would implement real cross-platform synchronization so the app could actually function across iOS and Android rather than being limited to a Figma prototype. Second, we would build out the facial detection feature more thoroughly, allowing the app to automatically suggest which contacts to share an album with based on who appears in the photos. Third, we would add a timeline or memories feature that surfaces past albums and memories at relevant times, similar to how Apple Photos and Google Photos show throwback memories. Finally, we would conduct additional rounds of usability testing with a larger and more diverse participant pool to validate our design changes and identify any remaining issues before development.